

UNIVERSITY OF THE ARMED FORCES - ESPE

Computer Science Department

Object Oriented Programming

36863

HomeWork-####

LASCANO, JORGE EDISON

October 14, 2026

Technician

```
Attributes:
      -echnicianId: String
      -name: String
      -specialty: String
      -isAvailable: bool
Methods:
      +assignRepair(order: RepairOrder)
      +updateLaborCost(cost: float)
      +getAssignedTasks(): List
```

WorkshopManager

```
Attributes:
      -managerId: String
      -username: String
      -accessLevel: int
Methods:
      +approveRefund()
      +viewMonthlyReport(report: MonthlyReport)
      +modifyInventoryThreshold(limit: int)
```

ExcelSpreadsheet

```
Attributes:
      -spreadsheetName: String
      -filePath: String
      -isStatic: bool
Methods:
      +manualDataEntry()
      +calculateCostsManually(): float
      +trackInventoryWithoutAlerts()
```

DiagnosticReport

```
Attributes:
      -reportId: String
      -hardwareChecklist: List
      -softwareChecklist: List
      -technicianNotes: String
Methods:
      +recordFindings()
      +attachToOrder(order: RepairOrder)
      +validateDiagnostic(): bool
```

Receipt

```
Attributes:
      -receiptNumber: String
      -issueDate: Date
```

-taxAmount: float

-totalPaid: float

Methods:
+generatePrintableFormat(): String
+sendEmailReceipt(customer: Customer)
+verifyPayment(): bool

InvoiceItem

Attributes:
-itemId: String
-description: String
-taxAmount: float
-unitPrice: float

Methods:
+calculateSubtotal(): float
+updateQuantity(qty: int)

Supplier

Attributes:

-supplierId: String

-companyName: String

-contactNumber: String

-suppliedParts: List

Methods:

+placeSparePartOrder(part: SparePart, qty: int)

+updateCatalog()

PurchaseOrder

Attributes:

-poNumber: String

-orderDate: Date

-deliveryStatus: String

Methods:

+addItem(part: SparePart, qty: int)

+receiveShipment()

AuditLog

Attributes:

-logId: String

-timestamp: Date

-actionType: String

-userReference: String

Methods:

+recordEvent()

+exportLogFile(): String

BackupManager

Attributes:

-backupId: String

-lastBackupDate: Date

-fileSize: float

Methods:

+generateDatabaseBackup()

+restoreDatabase()

TextFileLog

Attributes:

-fileName: String

-recordContent: String

-storageLocation: String

Methods:

+writePlainTextEntry()

+searchRecordManually(): String

+exportSummaryList()

PersistentDataBase

Attributes:
- -databaseName: String
- -connectionStatus: bool
- -recordCount: int

Methods:
- +storeServideRecord()
- +retrieveRecordData(): SystemRecord
- +consolidateMonthlyData()

NotificationSystem

Attributes:
- -alertThreshold: int
- -messageContent: String
- -isActive: bool

Methods:
- +triggerTextAlert()
- +displayStockWarning()
- +notifyCustomerStatus()

IntakeForm

Attributes:
- -formId: String
- -isStructured: bool
- -fieldsCompleted: bool

Methods:
- +captureDeviceInformation()
- +validateIntakeFields(): bool
- +generateInitialRecord()

StatusDropdownMenu

Attributes:
- -slectedOption: String
- -predefinedOptions: List
- -isVisible:bool

Methods:
- +updateActivePhase(newPhase: String)
- +renderStatusOptions()
- +reflectRealTimeProgress()

VisualCalendar

Attributes:
- -calendarId: String
- -selectedDate: Date
- -isInteractive: bool

Methods:
    +selectCompletationDate(): Date
    +displayTurnaroundTimeline()
    +setCustomerExcectation()

WarrantyCheckbox

Attributes:
    -checkboxId: String
    -isChecked: bool
    -validationStatus: String
Methods:
    +verifyWarrantValidity(): bool
    +documentIntakeCondition()
    +establishServiceBillingType()

PortalInterface

Atributes:
    -portalUrl: String
    -searchQueryInput: String
    -sessionActive: bool
Methods:
    +renderSearchScreen()
    +executeOrderLookup(orderNumber: String): RepairOrder
    +displayRealTimeStatus()

Customer

Attributes:
    - customerId : String
    - fullName : String
    - phoneNumber : String
    - email : String
Methods:
    + getContactDetails()
    + updateInformation(newEmail : String, newPhone : String)
    + requestServiceStatus(orderNumber : String) : RepairOrder

Device

Attributes:
    - serialNumber : String
    - deviceType : String
    - brand : String
    - model : String
    - reportedFaults : String
Methods:
    + getTechnicalProfile()
    + updateReportedFaults(newFaults : String)
    + getAssociatedCustomer() : Customer

RepairOrder

```
Attributes:
      - orderNumber : String
      - entryDate : Date
      - estimatedCompletionDate : Date
      - currentStatus : String
      - isValidUnderWarranty : bool
Methods:
      + updateOrderStatus(newStatus : String)
      + calculateTotalCost() : FinancialRecord
      + getDeviceDetails() : Device
```

SparePart

```
Attributes:
      - partId : String
      - partName : String
      - unitCost : float
      - stockQuantity : int
Methods:
      + checkStockLevel() : bool
      + updateStockQuantity(quantity : int)
      + isLowStock() : bool
```

FinancialRecord

```
Attributes:
      - recordId : String
      - laborCost : float
      - partsCost : float
      - totalCharge : float
Methods:
      + computeTotalCharge() : float
      - validateFinancialInput() : bool
      + getAssociatedOrder() : RepairOrder
```

SearchEngine

```
Attributes:
      - searchQuery : String
      - filterCriteria : String
Methods:
      + filterJobsByCustomerName(customerName : String) : List
      + filterJobsBySerialNumber(serialNumber : String) :
      RepairOrder
      + executeExternalOrderLookup(orderNumber : String) :
      RepairOrder
```

MonthlyReport

Attributes:

- reportMonth : int
- reportYear : int
- totalMoneyCollected : float
- completedRepairsCount : int

Methods:

+ compileMonthlyData()
+ exportToPlainText() : String

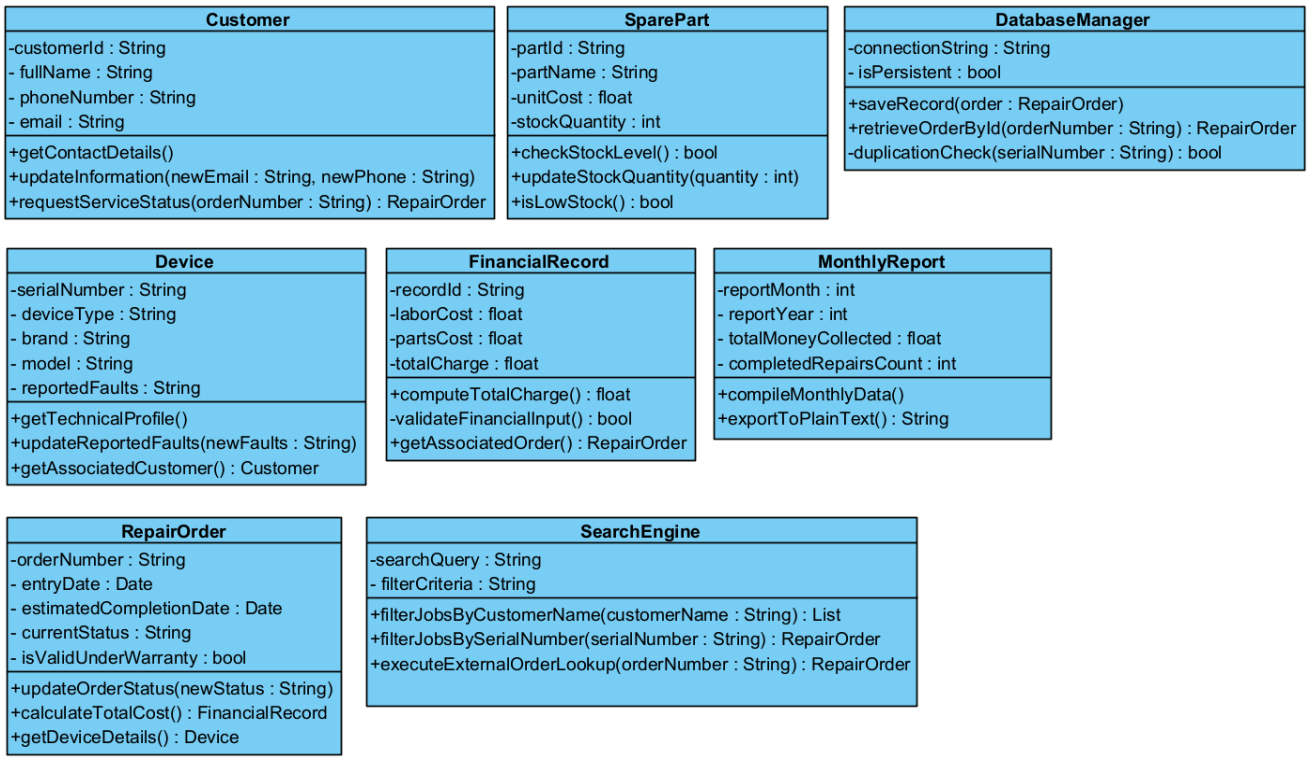
DatabaseManager

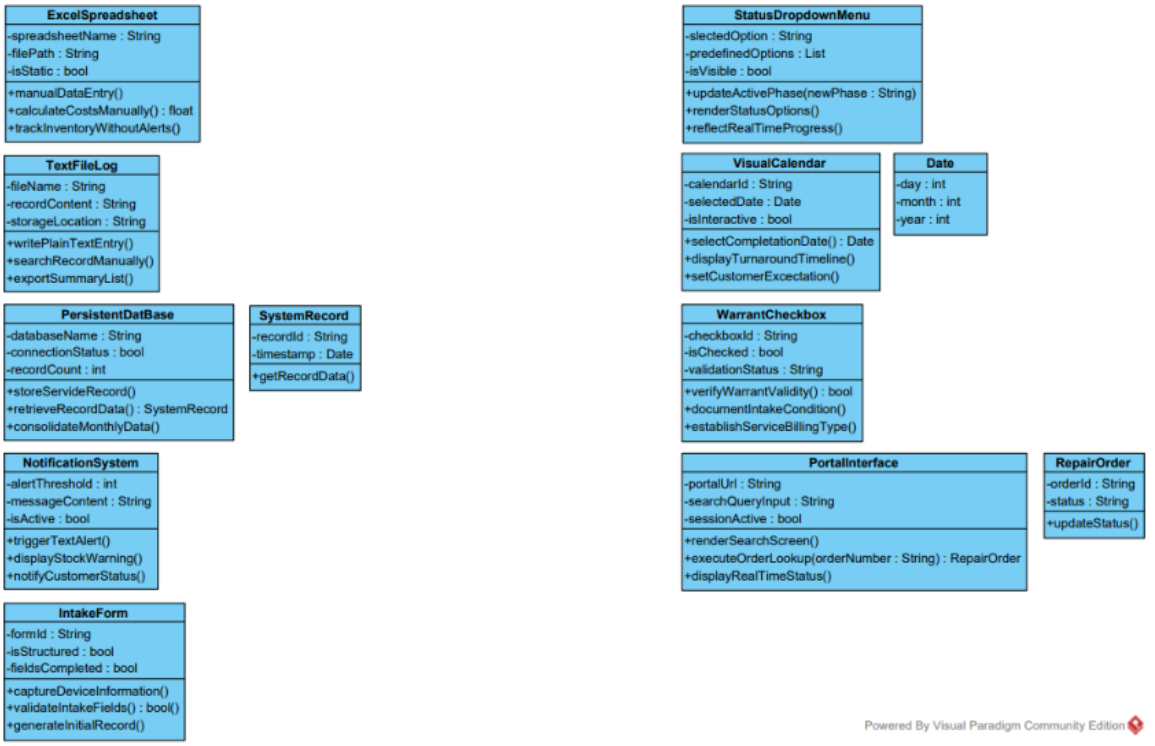
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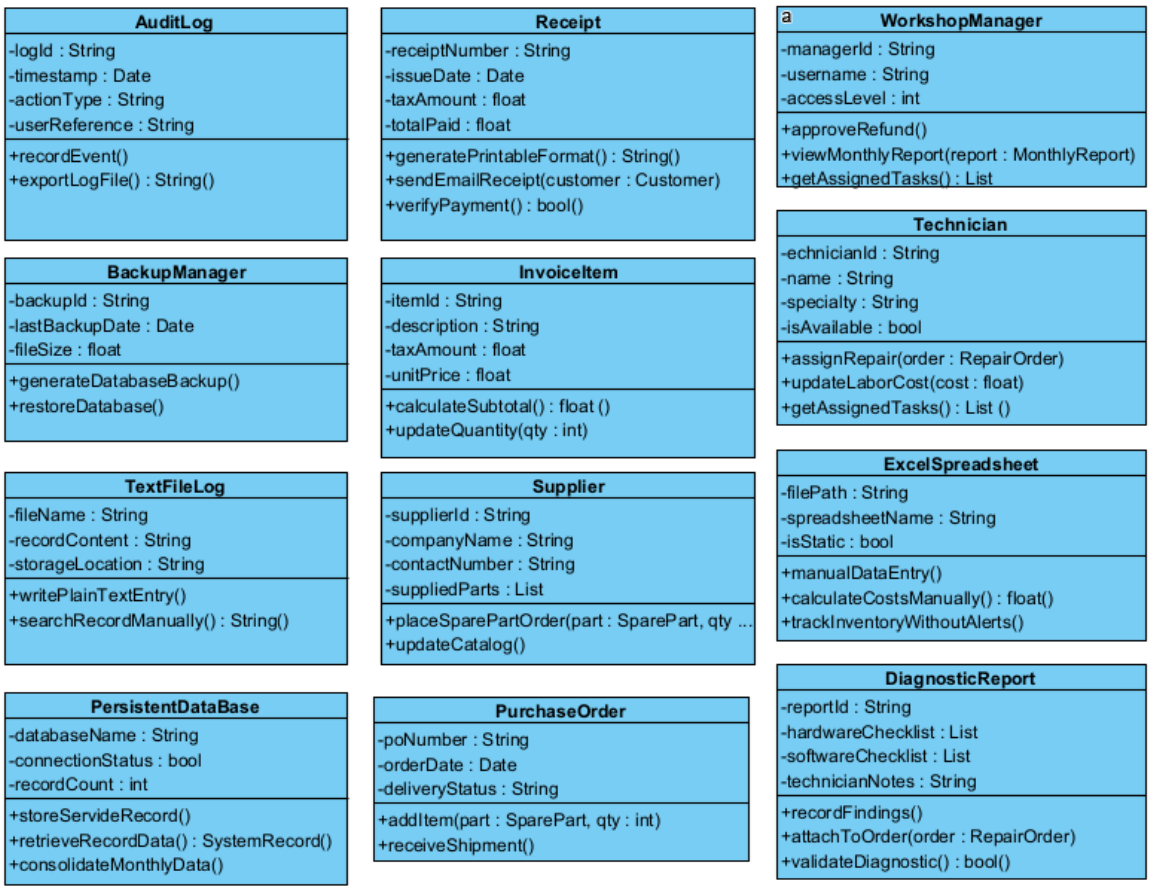
- connectionString : String
- isPersistent : bool

Methods:

+ saveRecord(order : RepairOrder)
+ retrieveOrderById(orderNumber : String) : RepairOrder
- duplicationCheck(serialNumber : String) : bool




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